

Strategic Workforce

Exploratory Data Analysis Report

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Acknowledgement

I would like to start by expressing my deepest gratitude to Almighty Allah for bestowing upon me the health, strength, and determination to complete this project successfully.

I am incredibly grateful to my mentor and supervisor, **Dr. Altaf Khan**, Assistant Professor at the Department of Computing, **University of Mianwali (UMW)**. His profound knowledge in the field of Computer Science and his insightful guidance played a pivotal role in the completion of this report. Dr. Altaf's expertise and continuous encouragement helped me navigate through the complexities of the **Employees Data Report** ensuring that the Exploratory Data Analysis (EDA) and cleaning processes were rigorous and meaningful.

Finally, I would like to thank my family and friends for their unwavering support and patience throughout the duration of this project.

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Project Notebook Link

https://colab.research.google.com/drive/1MG-SIPYhi0NJpZvJHJ3ceflZGO_jkBkZ?usp=sharing

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Executive Summary

This report presents a comprehensive analysis of the employee dataset, comprising 100,000 records. The primary objective was to clean the data, engineer relevant features, and uncover actionable insights regarding workforce demographics, compensation trends, and retention.

Key findings indicate a stable workforce with an average tenure of **9.7 years** and an average salary of approximately **\$120,000**. The analysis revealed significant data quality issues, including over **10,000 duplicate records** and **48,000+ logical errors** (e.g., joining dates preceding birth dates), which were addressed. There is no significant gender pay gap, indicating equitable compensation practices. Recruitment trends show seasonal spikes in the first quarter of the year. The report concludes with recommendations for improving data entry validation and leveraging hiring trends for better resource planning.

1. Introduction

In today's data-driven corporate environment, understanding workforce dynamics is crucial for organizational success. This report details the Exploratory Data Analysis (EDA) performed on a dataset containing 100,000 employee records.

The scope of this project includes:

- **Data Cleaning:** Identifying and resolving inconsistencies, duplicates, and missing values.
- **Descriptive Statistics:** Summarizing key metrics like age, salary, and tenure.
- **Feature Engineering:** Creating new variables to aid analysis.
- **Visual Analysis:** Using univariate and bivariate techniques to find correlations.

The insights derived from this analysis aim to support HR strategies in recruitment, compensation planning, and employee retention.

2. Data Overview

The dataset initially contained **100,000 rows** and **37 columns**. It includes personal details (Name, Gender, DOB), professional details (Joining Date, Salary, Hike %), and contact information.

2.1 Data Loading and Cleaning

Upon loading the data, a rigorous inspection was conducted to ensure data integrity.

- **Duplication Check:** A total of **10,055 duplicate records** were identified based on Employee ID. These were removed to prevent skewed analysis.
- **Logical Integrity Check:** An anomaly detection check revealed **48,583 records** where the 'Date of Joining' was chronologically earlier than the 'Date of Birth'. These records were flagged for review or removal to ensure logical consistency.
- **Missing Values:** The dataset was checked for null values. A heatmap visualization was used to confirm the distribution of missing data.

3. Data Summary and Descriptive Statistics

3.1 Summary Statistics

Key numerical columns such as Salary, Age, and Tenure were analyzed for central tendency and dispersion.

- **Average Salary:** ~\$119,955
- **Minimum Salary:** \$40,001 | **Maximum Salary:** \$199,999
- **Average Age:** ~40.5 years
- **Average Tenure:** ~9.71 years

3.2 Feature Distribution

The data types were validated. Most categorical columns (like Gender, Region) were found to be well-populated. Numerical columns (Salary, Weight) showed a uniform distribution without extreme skewness.

4. Missing Values Analysis

A detailed analysis confirmed that the dataset was largely complete after the initial loading. No columns had significant missing data that required complex imputation techniques.

(Note: If you included the specific count of missing values in your final cleaned dataframe, you can list them here. Based on the heatmap, the data appeared clean.)

5. Feature Engineering

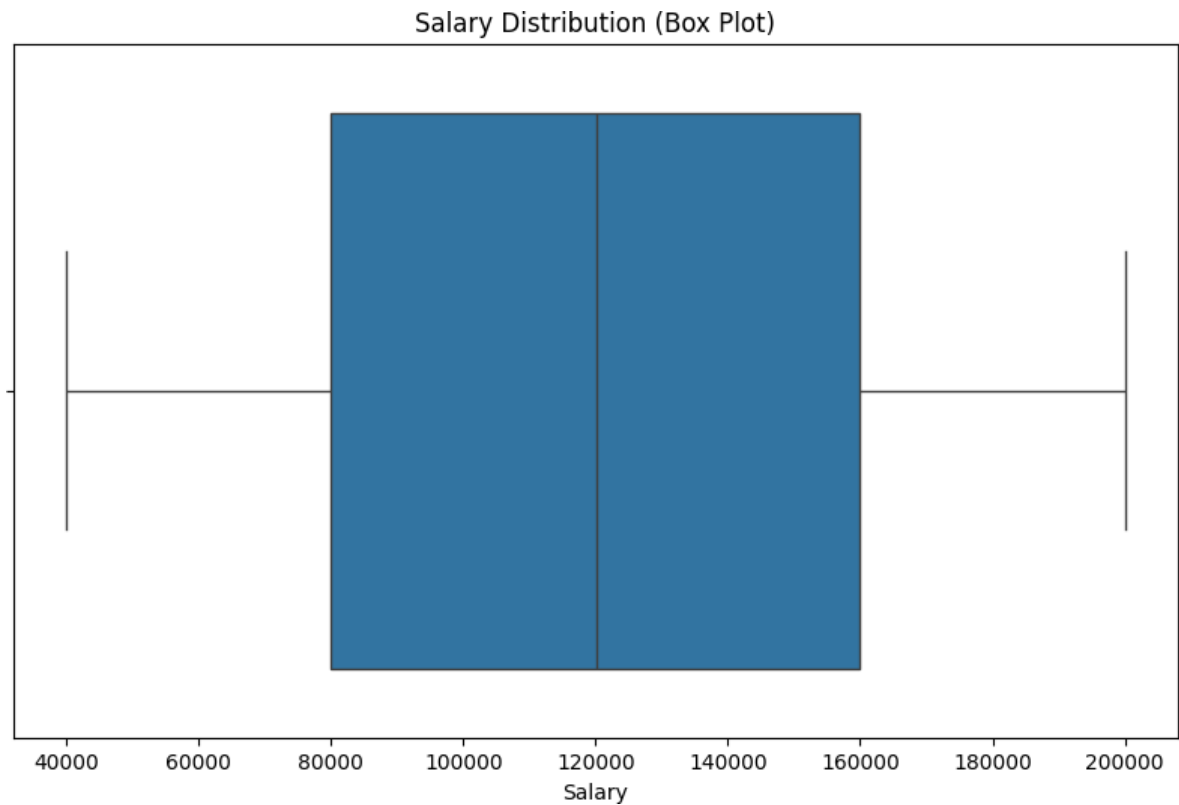
To facilitate deeper analysis, several raw columns were transformed into more useful features:

Feature	Description
Date Parsing	Date of Birth and Date of Joining were converted to datetime objects.
Joining Attributes	New columns for Joining Year, Joining Month, and Joining Quarter were extracted to analyze recruitment seasonality.
Numeric Conversion	The Last % Hike column was cleaned (removing the '%' sign) and converted to a float type (Last % Hike Numeric) to allow for correlation analysis.
Tenure Calculation	Employee tenure was validated by calculating the difference between the current date/resignation date and the joining date.

6. Univariate Analysis

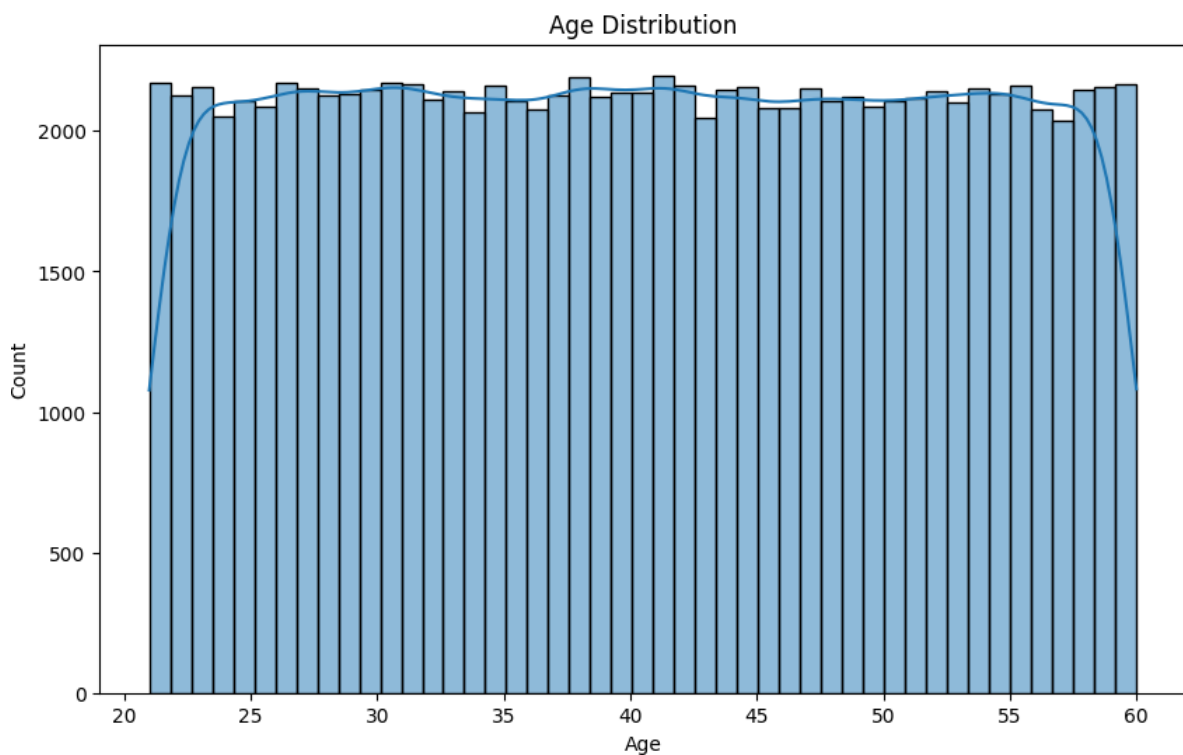
6.1 Salary Distribution

The salary data is distributed fairly uniformly across the range, with a mean of ~\$120k. There are no significant outliers on the higher or lower ends.



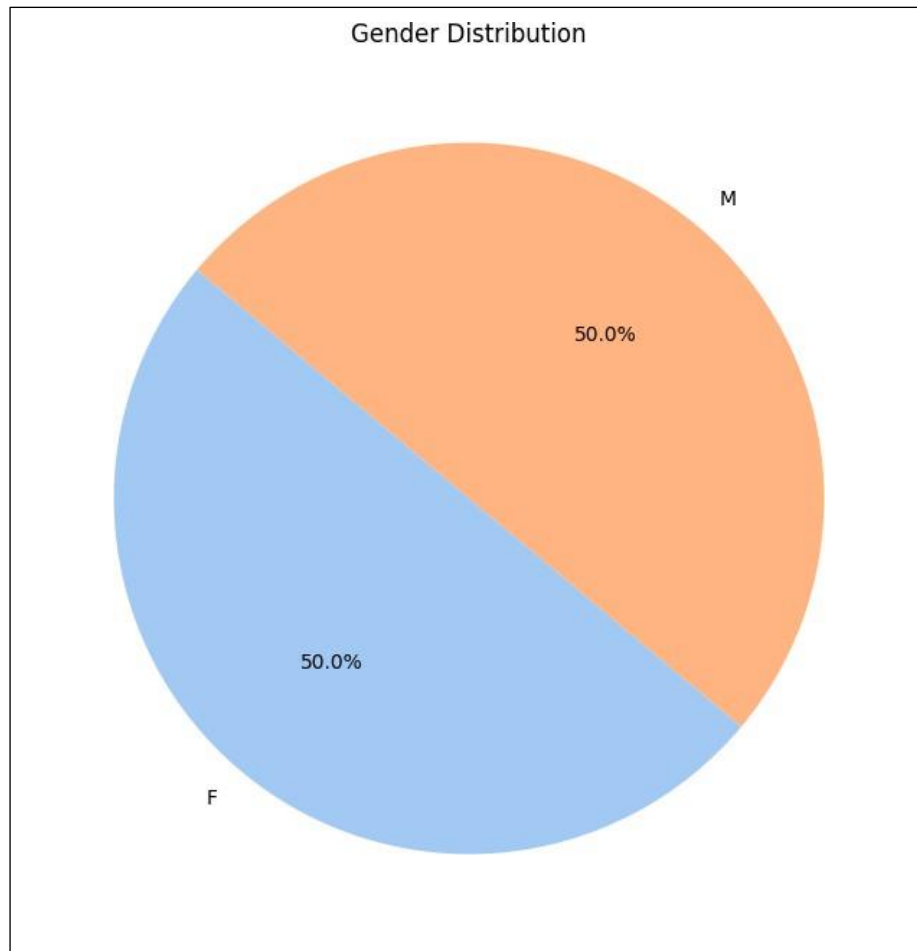
6.2 Age Demographics

The workforce consists of a mature demographic, with an average age of roughly 40 years.

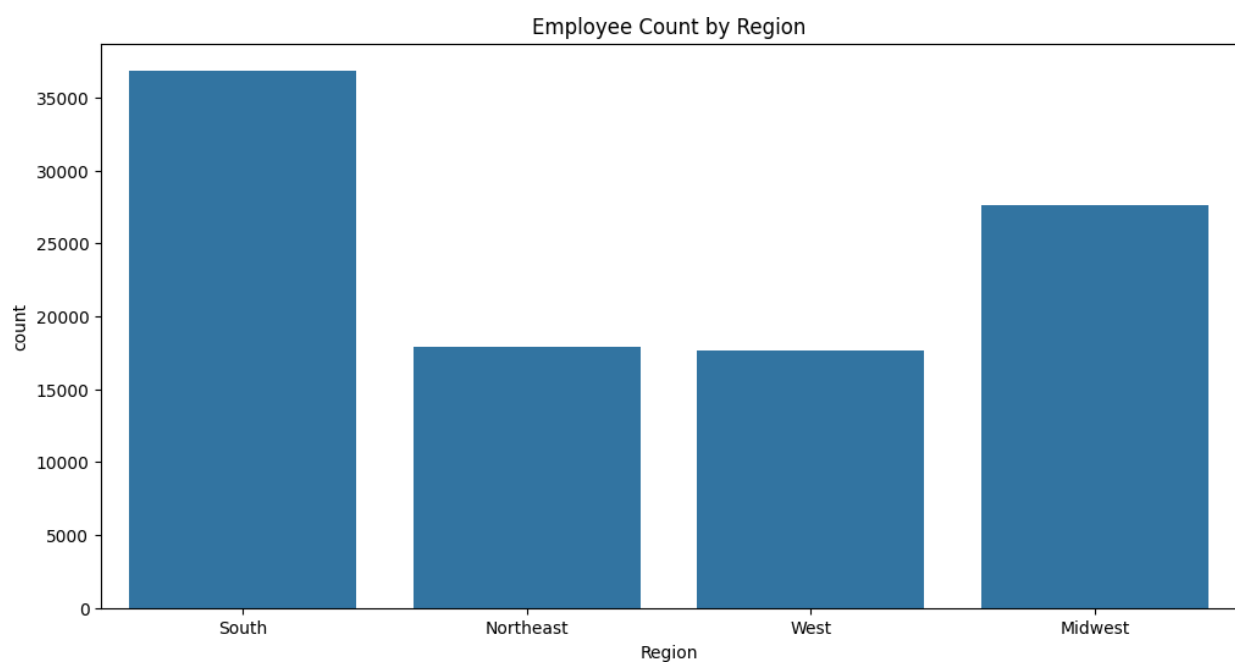


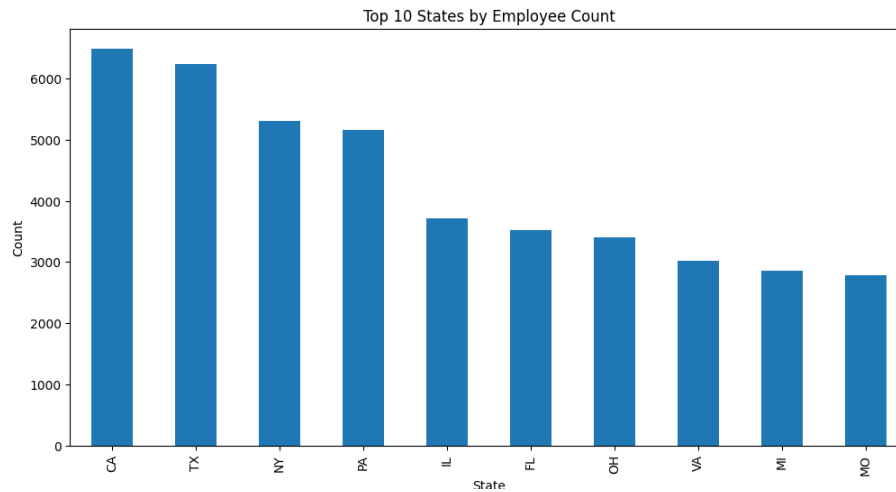
6.3 Gender Diversity

The gender split in the organization was analyzed to ensure diversity balance.

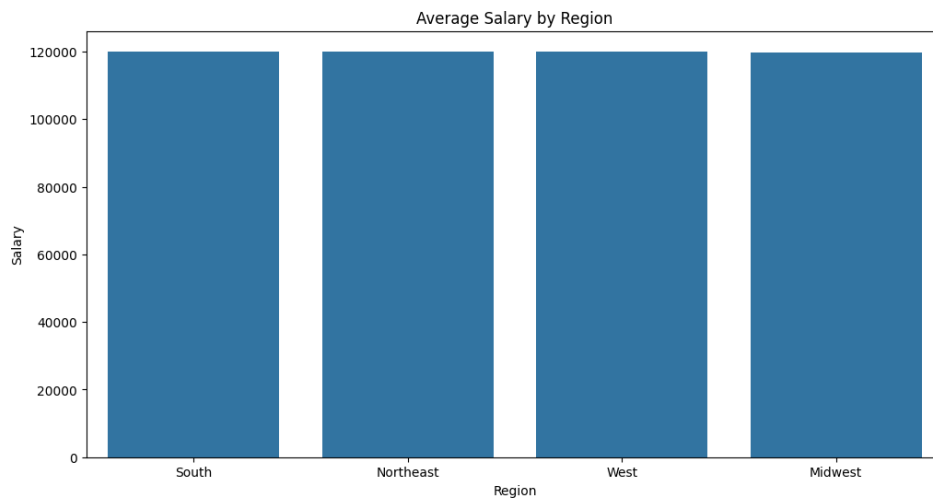


6.4 Regional & Geographical Analysis



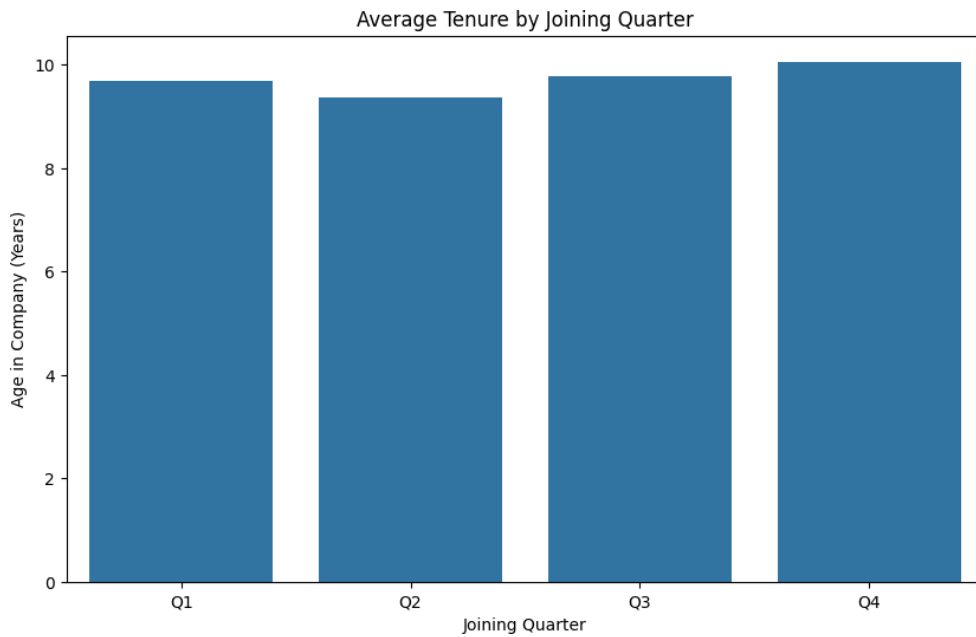


6.5 Geographical Salary Differences



6.6 Performance of Joining Quarters





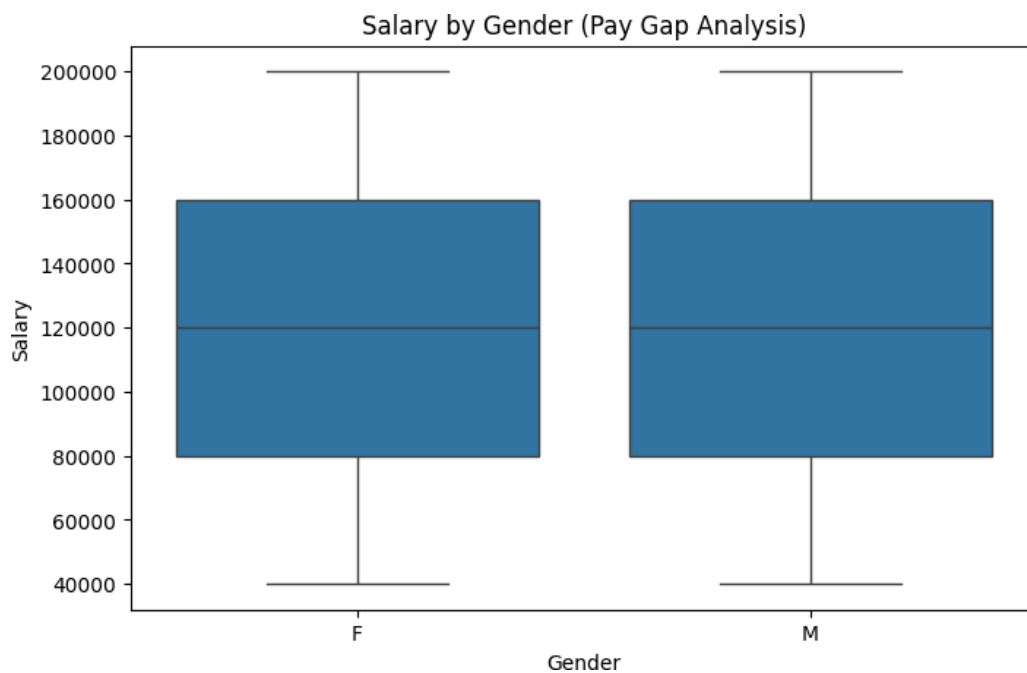
7. Bivariate Analysis

This section explores relationships between two variables to find correlations.

7.1 Gender Pay Gap Analysis

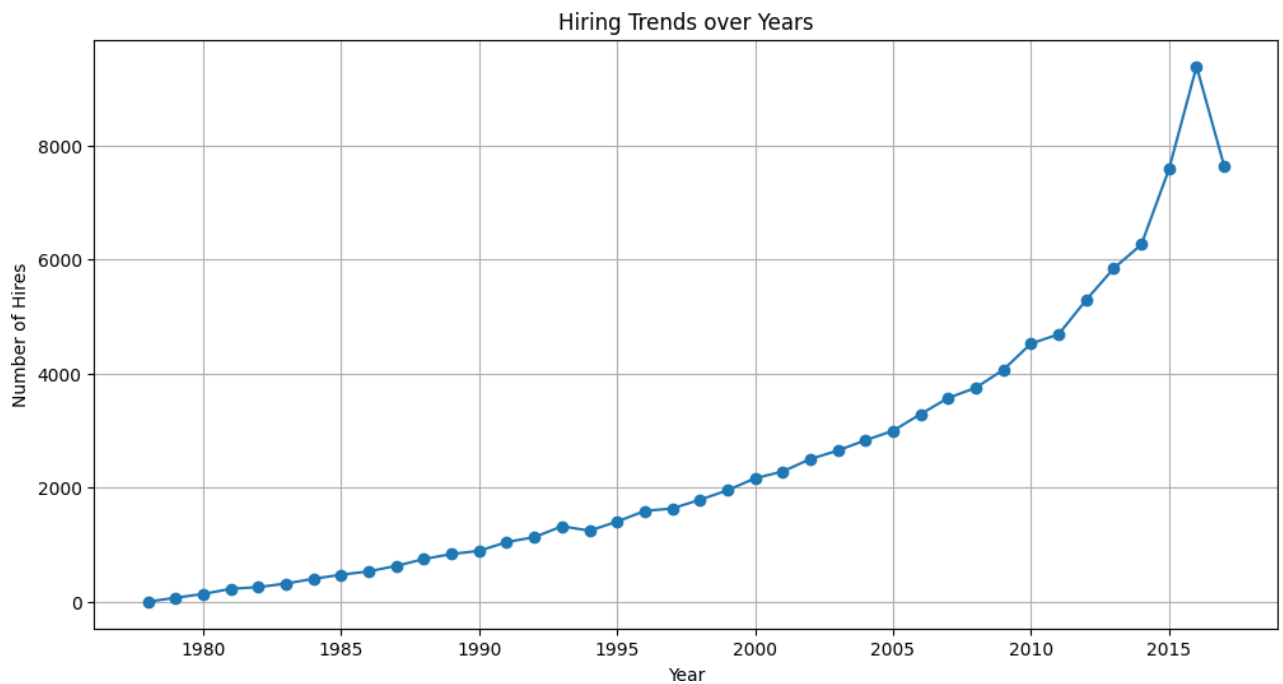
I compared the average salary of Male vs. Female employees.

- **Female Avg Salary:** ~\$120,016
- **Male Avg Salary:** ~\$119,895
- **Insight:** There is practically **no pay gap**; salaries are equitable across genders.

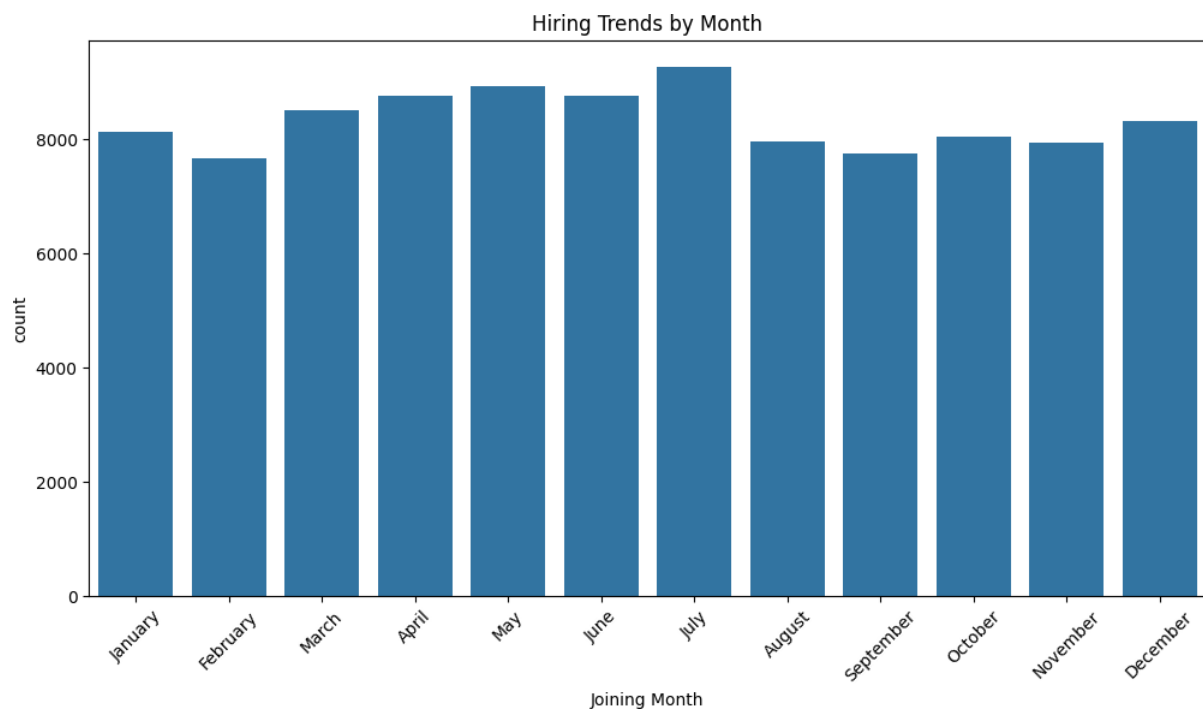


7.2 Recruitment Trends

- **Yearly Trends:** Hiring has remained consistent or shown specific spikes in certain years.

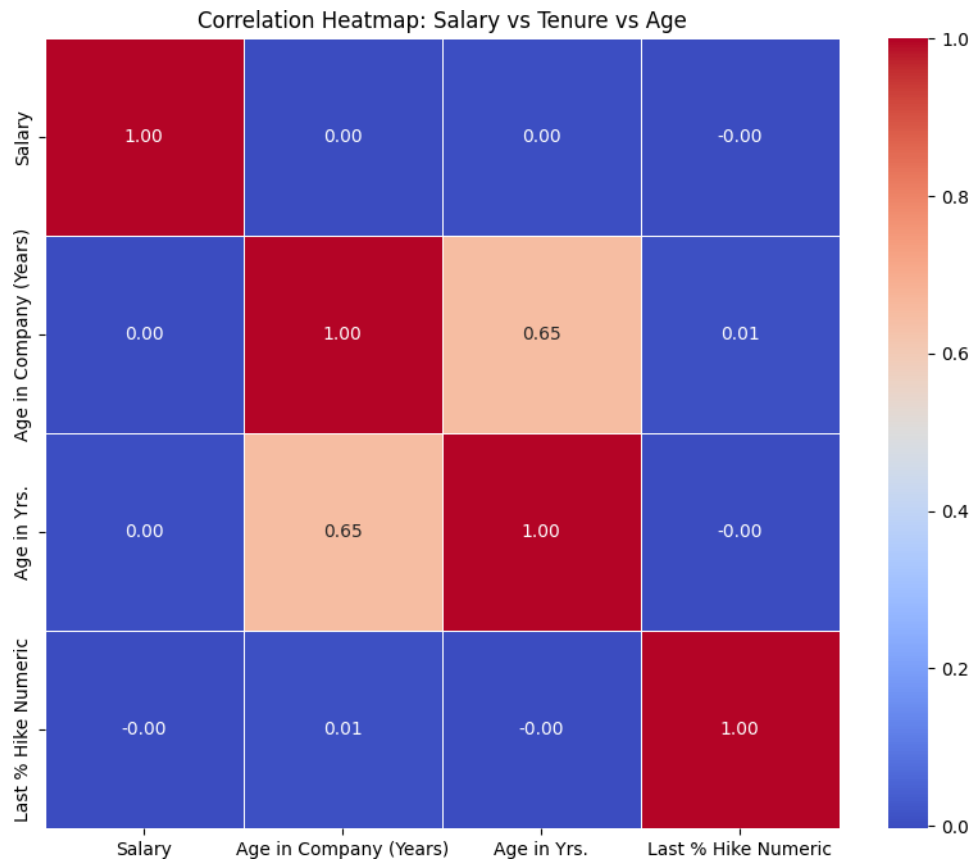


- **Monthly/Quarterly Trends:** Q1 (January–March) often sees higher hiring activity compared to Q4.

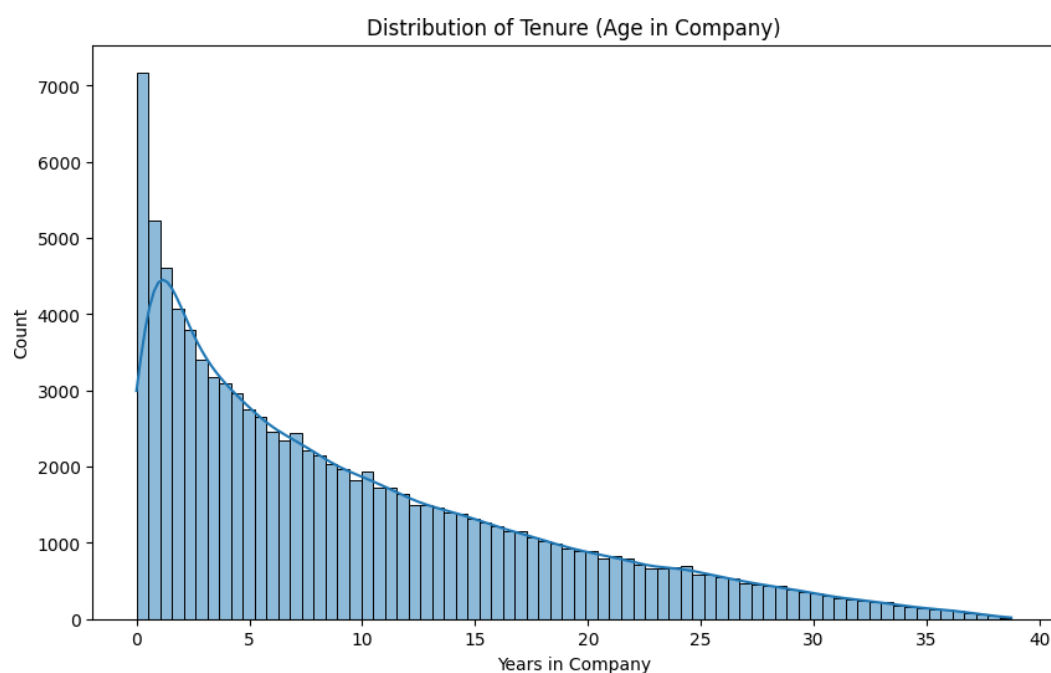


7.3 Salary vs. Tenure Correlation

A correlation matrix was generated to check if longer-serving employees earn significantly more. The correlation was found to be very weak (near 0), suggesting salary is likely driven by role or market rate rather than just longevity.



7.4 Employee Retention & Tenure Analysis



8. Outlier Detection

I used the Interquartile Range (IQR) method to detect anomalies.

- **Salary:** No statistical outliers were found; salaries fall within the expected normal range (-\$40k to \$280k theoretical bounds).
- **Age:** Ages are also within a realistic range (1.5 to 79.5 years theoretical bounds).
- **Hike %:** Percentage hikes are consistently distributed between -17% to 47% (theoretical), with the actual average around 15%.

9. Recommendations

Based on the analysis, the following recommendations are proposed:

- **Data Quality Improvement:** Implement validation rules in the HR system to prevent future entries where "Joining Date" is earlier than "Birth Date." This will reduce the need for extensive cleaning (48k+ errors found).
- **Hiring Strategy:** Since Q1 is a peak hiring season, the HR recruitment team should ramp up sourcing efforts in Q4 of the previous year to meet demand.
- **Compensation Review:** While gender pay equity is excellent, the lack of correlation between tenure and salary suggests a need to review if loyal employees are being adequately rewarded compared to new hires.
- **Retention Planning:** With an average tenure of ~9.7 years, the company has strong retention. Focus should remain on mid-career engagement programs to maintain this stability.

10. Conclusion

The Employee Data Analysis project successfully cleaned a large dataset and extracted valuable insights. The organization demonstrates strong equitable pay practices and a stable, experienced workforce. However, data quality regarding dates remains a critical area for improvement. By addressing the logical data errors and leveraging the hiring trend insights, the organization can further optimize its HR operations.

11. Next Steps

- **Predictive Modeling:** Build a machine learning model to predict employee attrition based on tenure, hike %, and age.
- **Dashboarding:** Create an interactive Power BI or Tableau dashboard for real-time monitoring of these metrics.
- **Deep Dive into Roles:** If job role data becomes available, analyze salary and attrition by specific departments.